

Assessment: MySQL Database (Joins & Nested Queries)

Scenario: Online Bookstore Management System

An online bookstore maintains information about customers, books, orders, publishers, and payments. The management wants to generate business reports to improve sales, identify loyal customers, and monitor inventory. As a database developer, you are required to write SQL queries using **Joins** and **Nested Queries** to retrieve meaningful information from the database.

Database Tables

- **Customers** (CustomerID, CustomerName, Email, City)
 - **Publishers** (PublisherID, PublisherName, Country)
 - **Books** (BookID, Title, Category, Price, PublisherID)
 - **Orders** (OrderID, CustomerID, OrderDate)
 - **OrderDetails** (OrderID, BookID, Quantity)
 - **Payments** (PaymentID, OrderID, PaymentMethod, Amount)
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Task 1 (5 Marks) – Joins

Write an SQL query to display the following information for every book that has been ordered:

- Customer Name
- Book Title
- Publisher Name
- Quantity Purchased
- Order Date

Use appropriate **INNER JOIN** statements to retrieve the required information.

Task 2 (5 Marks) – Nested Query

The management wants to reward premium customers.

Write an SQL query to find the names of customers who have placed **at least one order containing a book whose price is greater than the average price of all books**.

Use a **Nested Query** to solve the problem.

Task 3 (5 Marks) – Joins + Nested Query

The bookstore wants to identify its most valuable publishers.

Write an SQL query to display:

- Publisher Name
- Total Number of Books Sold

Only include publishers whose total books sold are greater than the average number of books sold by all publishers.

Use **JOINS** and a **Nested Query** together.

Task 4 (5 Marks) – Complex Business Report

Management wants to identify customers eligible for a **Gold Membership**.

Write an SQL query to display:

- Customer Name
- Total Number of Orders
- Total Amount Paid

Only display customers whose total payment is greater than the average payment made by all customers.

Use **Joins**, **GROUP BY**, and a **Nested Query** to generate the report.

Assessment Outcomes

By completing this assessment, students should be able to:

- Apply **INNER JOIN** to combine data from multiple related tables.
- Use **Nested Queries (Subqueries)** to solve real business problems.
- Analyze relational data using aggregate functions.
- Generate decision-making reports using SQL.
- Demonstrate problem-solving skills in a real-world database environment.